Juno Kim

PhD Student · Embodied AI · Robot Perception · Vision–Language–Action

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RESEARCH INTERESTS

Embodied AI, Robot Perception, Vision–Language–Action (VLA) models, 3D Scene Understanding, Robot Learning, and Sim2Real Transfer — with a focus on integrating explicit 3D perception with policy learning for robust manipulation and mobile robotics.

EDUCATION

Ph.D. in Interdisciplinary Program in Artificial Intelligence, Seoul National University 2022 – present
Advisor: Prof. Byoung-Tak Zhang · Biointelligence Lab (Expected: Feb 2028) GPA: 3.87 / 4.30

B.Eng. in Computer Convergence Software, Korea University 2018 – 2022
Minor: Artificial Intelligence GPA: 4.21 / 4.50

PUBLICATIONS

Underlined bold = author. * equal contribution. † corresponding author.

International Conferences

[1] **J. Kim***, Y. Park*, H.-J. Yoon*, B.-T. Zhang†. **Scene-Q: Confidence-Aware Coarse-to-Fine Querying of 3D Scenes with Selective VLM Reasoning**. *IROS 2026*.

[2] **J. Kim***, H.-J. Yoon*, Y. Park*, B.-T. Zhang†. **CDIS: Cross-Dimensional Class-Agnostic 3D Instance Segmentation via 2D Mask Tracking and 3D–2D Projection Merging**. *IROS 2025*.

[3] Y. Park*, H.-J. Yoon*, **J. Kim***, B.-T. Zhang†. **DA-Fusion: Deformable Attention-Based RGB-D Fusion Transformer for Unseen Object Instance Segmentation**. *ICRA 2025*.

[4] H.-J. Yoon*, **J. Kim***, Y. Park*, J.-K. Lee, B.-T. Zhang†. **Seg2Grasp: A Robust Modular Suction Grasping in Bin Picking**. *IROS 2024*.

[5] **J. Kim***, Y. Park*, H.-J. Yoon*, B.-T. Zhang†. **Ov-Map: Open-Vocabulary Zero-Shot 3D Instance Segmentation Map for Robots**. *IROS 2024*.

Workshops

[6] **J. Kim**, Y. Park, H.-J. Yoon, B.-T. Zhang. **Escape Is Not Success: Diagnosing Spatial-OOD Failures of Robot Action Interfaces**.
Everything Beneath the Policy Workshop, CoRL 2026.

[7] **J. Kim**, Y. Park, H.-J. Yoon, B.-T. Zhang. **Head2Wrist: Wrist-Camera Robot Policies from Human Head-Camera Demonstrations without a Scene Scan**.
The UMI Arena: From Handheld Devices to Real-World Policies Workshop, CoRL 2026.

[8] H.-J. Yoon, **J. Kim**, Y. Park, B.-T. Zhang. **Sim-to-Real Transfer of Generated Mobile-Manipulation Demonstrations Requires the Deployment Scene**.
Sim-to-Real-to-Field Workshop, CoRL 2026.

[9] H.-J. Yoon, **J. Kim**, Y. Park, B.-T. Zhang†. **Visual Perception-Based Assistive Mobile Robot System for Manipulation Tasks**. *Assistive Manipulation Workshop, ICRA 2023.*

Domestic Conferences

[10] **J. Kim**, B.-T. Zhang†. **Autonomous Recycling Robot System Using Heuristic Grasping Algorithms**. *KCC 2023*. Outstanding Paper Award

[11] H.-J. Yoon, Y. Park, **J. Kim**, Y. Yoo, B.-T. Zhang†. **Cashier-Free Counter System Using Artificial Intelligence with Robot Arm**. *KSC 2022*.

[12] Y. Yoo, H.-J. Yoon, **J. Kim**, Y. Park, B.-T. Zhang†. **Product Identification System Based on Image Feature Similarity for Logistics Automation**. *LOGIST 2022*.

PATENTS

Method and Apparatus for Segmenting an Instance of an Object KR, US
Patent Application Filed

Method and Apparatus for Constructing a Zero-Shot Object Map KR, US
Patent Application Filed

Method and Device for Controlling an Action for Picking an Object KR, US
Patent Application Filed

Method and Apparatus for Segmenting a 3D Instance of an Object KR
Patent Application Filed

RESEARCH EXPERIENCE

Graduate Researcher — Biointelligence Lab, Seoul National University 2022.9 – present
Advisor: Prof. Byoung-Tak Zhang

- Conduct research in robot perception, embodied AI, and Vision–Language–Action (VLA) models for manipulation and mobile robotics.
- Developed learning and perception methods for 3D instance segmentation, open-vocabulary scene understanding, and grasping in robotic systems.
- Proposed approaches for improving VLA robustness under visual distribution shift by integrating explicit perception with policy learning.
- Developed Real-to-Sim-to-Real pipelines for adapting pretrained VLA models to mobile manipulation using scene reconstruction and scalable trajectory generation.
- Built and evaluated deep learning and robotic systems in simulation and the real world using PyTorch, RGB-D perception, and robotic simulation platforms.

Research Intern — Korea AI Labs, Korea University 2021.12 – 2022.02

- Built a multi-modal model on top of KGAT and KGIN to predict and recommend popular items from Naver shopping data.

Research Intern — Machine Learning & Vision Lab, Korea University 2020.12 – 2021.09

- Reproduced *Diffusion Improves Graph Learning* and *Graph Contrastive Learning with Adaptive Augmentation*.
- Introduced node gradient as a new feature-importance factor and designed new augmentations based on it.

HONORS & AWARDS

Presidential Award (1st Place, Grand Prize) — ZEUS Industrial Robot ZERO Mission Challenge 2022
International Robot Contest (R-BIZ Challenge)

Champion (1st Place), DSPL & OPL Leagues — RoboCup@Home International AI Robot Competition 2023

Minister of Trade, Industry and Resources Award (1st Place, Grand Prize) — Humanoid AI Challenge 2026

Minister of Science and ICT Award (1st Place, Grand Prize) — AI Competition for Space Radio Disaster Prediction 2023

Head of LG AI Research Award (Excellence Award) — AI Co-Scientist Challenge Korea 2026

Director General of National Disaster Management Research Institute Award — Disaster Safety Data Hackathon 2024
National Disaster Management Research Institute

Grand Prize (1st Place) — Creative Autonomous Research Competition 2022, 2023
Interdisciplinary Program in Artificial Intelligence, Seoul National University

Outstanding Paper Award — Korea Computer Congress (KCC) 2023

Honorable Mention — Logistics Paper Competition 2022
Korean Society of Logistics Science & Technology

Dean's List — Korea University 2021
GPA 4.5 / 4.5 (Spring 2021) · Student of the Semester · Academic Excellence Award

National Excellent Student in Science & Engineering 2020 – 2021

SCHOLARSHIPS & FELLOWSHIPS

AI Fellowship Scholarship 2022 – 2025

National Excellent Student in Science & Engineering Full Scholarship 2020 – 2021

Academic Excellence Scholarship 2020

Social Service and Leadership Scholarship 2018 – 2019

TECHNICAL SKILLS

Programming	Python, C, C++
Frameworks	PyTorch, RGB-D perception, robotic simulation platforms
Domains	3D instance segmentation, open-vocabulary scene understanding, grasping, VLA, Sim2Real
Languages	English — TOEIC 950, TEPS 469

Last updated: October 2026